

# SIPG Haifa Bayport – Plan for Building Monthly Capital (K) Time Series

This report explains how to turn the two raw data files

- `sipg_haifa_bayport_CIP_RMB.tsv`
- `sipg_haifa_bayport_other_signals.tsv`

into **monthly** capital stock series **K<sub>t</sub>** for SIPG's Haifa Bayport, in **thousands of NIS**, that are comparable to the HPC and IPC series you already built.

The report is meant to be handed to another GPT chat whose job is to actually write and run the code. It therefore:

1. Documents what is in each raw file (conceptually / column-wise).
2. Defines the economic objects we ultimately want (annual and monthly K for Panel C and for the monthly port panel).
3. Gives a step-by-step transformation pipeline from raw TSVs → **annual Bayport K** → **monthly Bayport K**.
4. Spells out where external inputs / assumptions are needed (e.g., FX, depreciation rate, commissioning month).

Throughout, I'll assume that HPC and IPC K-series construction has already fixed certain conventions (units, depreciation approach). The idea is to mirror those conventions as closely as possible.

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## 1. Target: What we ultimately want

### 1.1 Annual Panel C targets

Panel C of your LaTeX table is:

**Panel C: Haifa – Bayport (SIPG), annual, in kNIS, with columns:**

- `Year`
- `Net PPE (EoY, kNIS)` – net physical capital stock at end of year
- `Gross investment (kNIS)` – Capex during year (gross additions to the capital stock)
- `Depreciation (kNIS)` – economic or accounting depreciation for the year
- `Disposals (kNIS)` – asset retirements / sales

We need these for **2018–2024**.

## 1.2 Monthly K-series targets

For the **monthly** K-series that will be merged with HPC and IPC, we ultimately want a file like:

- bayport\_K\_monthly.tsv with (at least) columns:
- year
- month
- date (e.g., YYYY-MM-01)
- owner (e.g., "SIPG" )
- K\_net\_kNIS – net capital stock at end of month
- I\_gross\_kNIS – gross investment during the month
- Dep\_kNIS – depreciation during the month
- Disposals\_kNIS – disposals during the month (likely 0 or negligible)

The monthly K-series should be consistent with the **annual Panel C values** in the sense that:

- Annual Gross investment  $\approx$  sum of monthly I\_gross\_kNIS within the year.
- Annual Depreciation  $\approx$  sum of monthly Dep\_kNIS within the year.
- Annual Net PPE (EoY)  $\approx$  K\_net\_kNIS at December of that year.

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## 2. What is in the two raw TSV files

### 2.1 sipg\_haifa\_bayport\_CIP\_RMB.tsv

This file is the **central input**. Conceptually, it is the cleaned and standardized version of all Bayport-related items extracted from the SIPG 2018–2024 reports, focusing on **construction-in-progress (CIP)** and Bayport-specific projects.

Columns (exact names may differ slightly; the other chat should inspect the header to confirm):

- year – reporting year (2018, 2019, ..., 2024)
- project\_name\_en – English project label (e.g., "Haifa Bayport port engineering project", "Bayport rail area fire-cannon system project" )
- project\_name\_cn – Chinese project label
- category – e.g. "main\_Bayport\_CIP" for the big terminal build-out, "Bayport\_fire\_cannon\_CIP" for the 2024 follow-on project, etc.
- budget\_rmb – total budget for that project, in RMB
- cip\_opening\_balance\_rmb – CIP book value at the **start** of the year, in RMB
- cip\_additions\_rmb – additions to CIP during the year (construction expenditures capitalized), in RMB
- transfers\_to\_PPE\_rmb – amount of CIP transferred to fixed assets (PPE) during the year, in RMB
- transfers\_to\_intangibles\_rmb – amount transferred to intangible or long-term deferred assets, if any
- other\_decreases\_rmb – other CIP decreases (write-offs, etc.), in RMB
- cip\_closing\_balance\_rmb – CIP book value at **end** of year, in RMB

- `progress_pct` – cumulative investment as a % of budget
- `cumulative_capitalized_interest_rmb` – total interest capitalized to date for this project (if present)
- `current_period_capitalized_interest_rmb` – interest capitalized in the current year
- `interest_cap_rate_pct` – capitalized interest rate (if present)
- `funding_source_en` / `funding_source_cn` – e.g., loans vs. self-financed
- `source_doc` / `source_page_note` – provenance in the SIPG reports

Key rows we care about:

- 2018–2021: rows for the **main Bayport terminal build**.
- 2024: row for the **Bayport rail-area fire-cannon CIP project**.

For the main Bayport project, CIP dynamics follow:

$$CIP_y^{close} = CIP_y^{open} + additions_y + capint_y - transfers\ to\ PPE_y - transfers\ to\ intangibles_y - other\ decreases_y.$$

For the Bayport main project, 2018–2021, the CIP table gives you:

- 2018: small starting balance, small additions, no transfers to PPE yet.
- 2019–2020: large additions, CIP builds up.
- 2021: very large transfers to PPE, CIP closing balance drops sharply (project substantially completed).
- 2022 onward: main project likely has zero or negligible CIP in this table (terminal is built), while small Bayport follow-on project appears in 2024.

## 2.2 `sipg_haifa_bayport_other_signals.tsv`

This file collects **other Bayport-related signals**, mostly used for validation and calibration, not for the core CIP arithmetic.

Examples of rows / columns include:

- **VAT receivable:**
  - Yearly amounts of `Bayport VAT to be refunded` in RMB, from “Other current assets” notes.
- **Masada EPC contract and payables:**
  - USD contract amount for the EPC contract; cumulative USD payments / RMB equivalents.
  - Related-party payables to Masada in RMB.
- **Registered capital and equity commitments:**
  - NIS registered capital and paid-in capital of SIPG Bayport Terminal Co., Ltd.
- **Guarantees:**
  - USD guarantee limit for Bayport project financing.
- **Foreign-currency monetary items:**
  - NIS-denominated cash / receivables / payables, with closing RMB conversion rates.
- **Tax rate and concession horizon:**
  - Israeli corporate income tax rate (e.g., 23%).
  - Concession lease expiry year (e.g., 2046).

For the **monthly K construction**, these signals matter mostly as **cross-checks and calibration aids**, and as a place to pull:

- The concession end year (to bound economic life).
- Some sense of timing (e.g., when operations started, from narrative notes if present).
- Rough FX rate anchors via NIS monetary items.

The **core quantitative driver of Bayport physical K** is still `sipg_haifa_bayport_CIP_RMB.tsv`.

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### 3. Step 1 – Build an annual Bayport CAPEX and stock series in RMB

The first step for the other GPT chat is to turn the CIP table into a **single annual panel** for the Bayport main project (plus the small 2024 project), in RMB.

#### 3.1 Filter to Bayport-relevant rows

1. Load `sipg_haifa_bayport_CIP_RMB.tsv` into a DataFrame.
2. Filter to rows where `category` indicates Bayport:
3. `category == "main_Bayport_CIP"` for the 2018–2021 construction-phase project.
4. `category == "Bayport_fire_cannon_CIP"` (or similar) for the 2024 follow-on project.

You can work with these two subsets separately and later combine them.

#### 3.2 Build an annual panel for the main Bayport project

For the **main Bayport project (2018–2021)**, construct an annual panel with columns:

- `year`
- `cip_open_rmb`
- `cip_close_rmb`
- `cip_additions_rmb`
- `cap_int_rmb` – current-period capitalized interest (if non-missing)
- `transfers_to_PPE_rmb`
- `transfers_to_intangibles_rmb`
- `other_decreases_rmb`

Use the values directly from the CIP rows (there is only one row per year for this project in your cleaned table, so you can just select by year).

Then, define **annual investment components** for Bayport in RMB:

- **Gross construction spending in year  $y$  (cash concept)**

Let

$$I_y^{\text{construct}} \equiv \text{cip\_additions\_rmb}_y + \text{cap\_int\_rmb}_y.$$

This is the **amount of new Bayport capital formation** in year  $y$ , viewed from the SIPG group's books (including capitalized interest as part of asset cost).

- **Assets put into service (completed works)**

Let

$$T_y^{\text{PPE}} \equiv \text{transfers\_to\_PPE\_rmb}_y,$$

which is the value of completed works that have been moved out of CIP and into PPE in year  $y$ .

- **Other exits**

Let

$$E_y^{\text{other}} \equiv \text{transfers\_to\_intangibles\_rmb}_y + \text{other\_decreases\_rmb}_y.$$

CIP should satisfy the accounting identity above; the other chat should sanity-check this year by year.

### 3.3 Define an annual gross capital stock at cost in RMB

We now want an annual time series for the **gross capital stock at cost** for Bayport.

Since the Bayport terminal did not exist before 2018 in SIPG's consolidated books, set:

- $\boxed{\text{K\_gross\_RMB\_2017} = 0}.$

Then define a **cumulative completed-works stock**  $\boxed{\text{K\_completed\_RMB\_y}}$ :

$$K_y^{\text{completed}} = \sum_{s \leq y} T_s^{\text{PPE}}.$$

This measures the total cost (in RMB) of assets that have been **transferred out of CIP into PPE** by year  $y$ .

If you want a measure of **total Bayport capital in place including unfinished works**, you can define:

$$K_y^{\text{gross}} = K_y^{\text{completed}} + \text{cip\_close\_rmb}_y.$$

This  $\boxed{\text{K\_gross\_RMB\_y}}$  can be thought of as "all resources sunk into Bayport infrastructure so far," regardless of whether they are still in CIP or already in PPE.

For the **annual Panel C**, you have two options:

- **Option A (simpler):** Use gross capital in place  $\boxed{\text{K\_gross\_RMB\_y}}$  as the basis for net PPE (and later adjust for depreciation).

- **Option B (stricter):** Use only `K_completed_RMB_y` as the productive base (only completed works are depreciated), treat CIP separately. For Panel C, you may prefer this, but the difference in the early years is not huge once the project is largely completed.

The other chat should implement both, label them clearly, and you can decide which to plug into Panel C.

### 3.4 Incorporate the 2024 Bayport follow-on project

The 2024 file has a separate CIP row for the **Bayport rail-area fire-cannon system project**. For this project:

- Treat it as **new Bayport investment** starting in 2024.
- It has its own `budget_rmb`, `cip_open_rmb` (likely 0), `cip_additions_rmb`, `cip_close_rmb`, etc.

To integrate this into the Bayport K series:

1. Compute `I^{\text{fire}}_y`, `T^{\text{fire}}_y`, etc. as above.
2. Add them to the main Bayport project totals for 2024:
3. `I_total_RMB_2024 = I^{\text{main}}_{2024} + I^{\text{fire}}_{2024}` (main may be 0 by 2024).
4. `T_total_RMB_2024 = T^{\text{main}}_{2024} + T^{\text{fire}}_{2024}`.
5. Update `K_completed_RMB_2024` and `K_gross_RMB_2024` accordingly.

## 4. Step 2 – Convert annual Bayport series from RMB to NIS

Your HPC and IPC K panels are in **thousands of NIS**. Bayport is currently in **RMB** in our cleaned tables. We need an FX bridge.

### 4.1 Build/obtain an FX table

Create a small FX table, e.g. `fx_rmb_nis.tsv`, with at least:

- `year`
- `fx_avg_nis_per_rmb` – annual average NIS per RMB
- `fx_eoy_nis_per_rmb` – end-of-year NIS per RMB (optional)
- `source` – e.g., Bank of Israel, IMF, etc.

The **other GPT chat** (or you) will need to pull these FX rates from an external source.

For capital stock and investment, **annual average FX** is usually more appropriate, because investment happens throughout the year. For EoY net PPE, you could use either:

- annual average FX, for consistency with HPC/IPC K; or
- closing FX, if you want to mimic SIPG book values more closely.

Pick one rule and apply it consistently.

## 4.2 Convert the annual RMB series to NIS

For each year  $y$ , once the FX table is merged, compute:

- $I\_total\_NIS\_y = I\_total\_RMB\_y * fx\_avg\_nis\_per\_rmb\_y$ .
- $T\_total\_NIS\_y = T\_total\_RMB\_y * fx\_avg\_nis\_per\_rmb\_y$ .
- $K\_completed\_NIS\_y = K\_completed\_RMB\_y * fx\_avg\_nis\_per\_rmb\_y$ .
- $K\_gross\_NIS\_y = K\_gross\_RMB\_y * fx\_avg\_nis\_per\_rmb\_y$ .

Convert to **thousands of NIS** for the Panel C table by dividing by 1,000:

- $I\_total\_kNIS\_y = I\_total\_NIS\_y / 1\_000$ .
- etc.

These NIS figures will later anchor both:

- the **annual Panel C** entries; and
- the **monthly PIM** (perpetual inventory method) construction.

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## 5. Step 3 – Define annual net PPE, gross investment, depreciation, disposals

With NIS-valued annual Bayport series in hand, we now need to decide how to define the Panel C variables.

### 5.1 Gross investment

The most transparent choice is:

$$\text{Gross investment}_y \equiv I_{kNIS,y}^{\text{total}}$$

where  $I\_total\_kNIS\_y$  is the NIS value of total construction spending plus capitalized interest in year  $y$ . This is the **broadest** measure of annual Bayport investment.

If you prefer to treat capitalized interest separately, you can instead set:

$$\text{Gross investment}_y = (cip\_additions\_rmb_y \times fx) / 1,000,$$

and keep a separate note or column for capitalized interest.

### 5.2 Net PPE at end of year

We need a **net** capital stock, which means we have to introduce **depreciation**. SIPG does not provide Bayport-specific PPE and depreciation flows at the project level in the group notes, so we must impose assumptions, parallel to your approach for IPC/HPC where you used a PIM with assumed service lives.

### 5.2.1 Choose a service life and depreciation rate

Let:

- $L_{\text{bayport}}$  = assumed economic life (years) of the Bayport infrastructure (e.g., 40–50 years), possibly informed by the **concession lease end year** (2046) from  $\text{other\_signals}$  and the main construction completion year (2021/2022).
- $\delta_{\text{annual}} = 1 / L_{\text{bayport}}$  under straight-line depreciation.

Alternatively, you can pick  $\delta_{\text{annual}}$  to match your **HPC/IPC depreciation conventions** for port infrastructure.

### 5.2.2 Choose when depreciation starts

We must decide in which year the Bayport assets begin depreciating.

- Use the year when the terminal **enters operation** (earliest year with meaningful throughput / revenue). In practice that's around 2021–2022. You can pick either:
- $\text{year\_dep\_start} = \text{first year when major transfers to PPE occurred}$  (likely 2021); or
- $\text{year\_dep\_start} = \text{first full year of operation}$  (likely 2022).

In the model:

- For  $y < \text{year\_dep\_start}$ , **no depreciation** is recorded; the entire gross stock remains undepreciated (as construction in progress and newly completed assets that are not yet used).
- For  $y \geq \text{year\_dep\_start}$ , depreciation is calculated on the stock of **productive capital**.

### 5.2.3 Perpetual inventory recursion at the annual level

Define  $K_{\text{net\_kNIS\_y}}$  as the **net capital stock at end of year y**. Then a standard PIM recursion is:

$$K_y^{\text{net}} = (1 - \delta_{\text{annual}})K_{y-1}^{\text{net}} + I_y^{\text{productive}} - D_y^{\text{disposals}}.$$

Where:

- $I_{\text{productive\_y}}$  = the share of **gross investment that actually enters productive capital** in year y.
- Simplest: take  $I_{\text{productive\_y}} = T_{\text{total\_kNIS\_y}}$ , i.e. only completed works transferred to PPE begin depreciating.
- More aggressive: take  $I_{\text{productive\_y}} = I_{\text{total\_kNIS\_y}}$ , i.e., treat all construction spending as capital in place from day one.
- $D_{\text{disposals\_y}}$  = annual disposals. For Bayport, the SIPG notes do not show any early disposals, so **we can safely set this to 0** for now.

To link back to gross stock  $K_{\text{gross\_kNIS\_y}}$ , you can also define net stock as:

$$K_y^{\text{net}} = K_y^{\text{gross}} - \text{cumulated depreciation}_y.$$



The other chat can implement the PIM in any of these equivalent ways; the important thing is to adopt a clear rule and then ensure the annual values are coherent.

### 5.2.4 Extract annual depreciation and disposals

Once `K_net_kNIS_y` is defined via PIM, we can back out the annual **depreciation** and store it for Panel C:

$$\text{Depreciation}_y = K_{y-1}^{\text{net}} + I_y^{\text{productive}} - D_y^{\text{disposals}} - K_y^{\text{net}}.$$

- For Bayport, set `D_disposals_y = 0` unless you find explicit disposal events.
- Store `Depreciation_y` in kNIS.

### 5.3 Fill Panel C once the annual series is built

After the PIM is implemented with chosen assumptions, the other chat can construct a table:

- `Year`
- `Net PPE (EoY, kNIS) = K_net_kNIS_y`
- `Gross investment (kNIS) = I_total_kNIS_y` (or your preferred definition)
- `Depreciation (kNIS) = Depreciation_y`
- `Disposals (kNIS) = D_disposals_y` (likely all zeros)

This directly populates Panel C.

## 6. Step 4 – Build monthly Bayport K-series using PIM

Once annual NIS-level series are defined, the next step is to distribute them over months and run a **monthly PIM**.

### 6.1 Framework

We want monthly net capital `K_t` that satisfies:

$$K_t = (1 - \delta_{\text{month}})K_{t-1} + I_t - D_t,$$

where:

- `t` indexes months (e.g., 2018-01, 2018-02, ..., 2024-12).
- `δ_month` is the **monthly depreciation rate**.
- `I_t` is monthly investment (in kNIS).
- `D_t` is monthly disposals (set to 0 unless you have an explicit disposal month).

### 6.2 Choose `δ_month`

Given an annual depreciation rate `δ_annual`, define:

$$\delta_{\text{month}} = 1 - (1 - \delta_{\text{annual}})^{1/12}.$$

This ensures that applying  $\delta_{\text{month}}$  over 12 months is equivalent to one application of  $\delta_{\text{annual}}$ .

### 6.3 Distribute annual gross investment across months

For each year  $y$ , we know  $\text{Gross\_investment}_y$  (in kNIS). We need to decide how to assign this to months.

Without detailed intra-year data, the **simplest and transparent rule** is **uniform distribution within year**:

$$I_t = \frac{\text{Gross investment}_y}{12} \quad \text{for all months } t \text{ in year } y.$$

More sophisticated patterns (e.g., front-loaded during construction years, or aligned with known milestone dates) can be added later, but uniform is a defensible starting point and matches what you probably did for IPC/HPC when you lacked monthly capex detail.

### 6.4 Decide the construction start month and first depreciation month

We need two timelines:

1. **Construction start month**: earliest month  $t$  with positive  $I_t$ .
2. Set this to 2018-01 (January 2018), since 2018 is the first year with Bayport CIP.
3. This is equivalent to spreading 2018 gross investment evenly from Jan–Dec 2018.
4. **Depreciation start month**: earliest month when Bayport assets are treated as **productive**.
5. For example, if you choose  $\text{year\_dep\_start} = 2022$ , set  $\text{dep\_start\_month} = 2022-01$ .
6. Before that month, you can choose:
  - either to set  $\delta_{\text{month}} = 0$ , so no depreciation is taken; or
  - to depreciate only the share of completed works after they are transferred to PPE (see next substep for a more nuanced version).

The other chat should implement a switch: before  $\text{dep\_start\_month}$ , depreciation is either zero or applied only to a subset of capital.

### 6.5 Level alignment between annual and monthly series

To ensure consistency between annual and monthly series:

1. Use the monthly PIM recursion to generate  $K_t$  from 2018-01 to 2024-12 with the uniform  $I_t$  schedule and chosen  $\delta_{\text{month}}$  and  $\text{dep\_start\_month}$ .
2. For each year  $y$ , look at  $K_t$  in December (e.g.,  $K_{\{y-12\}}$ ) and compare with the annual target  $K_{\text{net\_kNIS}_y}$  from the annual PIM.
3. If the difference is small, you can accept it as rounding error.

4. If there are non-trivial discrepancies, the other chat can **rescale the within-year investment profile** so that:
5. The sum of monthly  $I_t$  in year  $y$  still equals  $\text{Gross\_investment}_y$ , and
6. The December  $K_t$  matches  $K_{\text{net\_kNIS}_y}$  exactly.

A simple way to do this year-by-year alignment is:

- Run monthly PIM with a provisional  $I_t$  profile.
- Compute  $K^{\{\text{provisional}\}}_{y, \text{Dec}}$ .
- Compute a scaling factor  $\lambda_y = K^{\{\text{target}\}}_{y, \text{Dec}} / K^{\{\text{provisional}\}}_{y, \text{Dec}}$ .
- Multiply monthly  $I_t$  in year  $y$  by  $\lambda_y$  and rerun the PIM.

This ensures the monthly series exactly matches the annual targets by construction.

## 6.6 Disposal treatment

The SIPG notes do not show Bayport-specific disposals in the early years. So, only if you later identify a disposal event (e.g., a small Bayport asset being scrapped) should the other chat introduce a monthly  $D_t$ . Otherwise, set  $D_t = 0$  for all  $t$ , and Panel C disposals will be 0 as well.

## 6.7 Output format for bayport\_K\_monthly.tsv

Once the monthly series is built, have the other chat write out a clean TSV with columns like:

- `date` (YYYY-MM-01)
- `year`
- `month`
- `owner` (string, "SIPG")
- `K_net_kNIS`
- `I_gross_kNIS`
- `Dep_kNIS`
- `Disposals_kNIS`
- `notes` (optional, e.g., which version of  $\delta$ , `dep_start`, and `FX` were used)

This file is what you will merge into your **Haifa-wide monthly K panel** alongside HPC and IPC.

## 7. How to use sipg\_haifa\_bayport\_other\_signals.tsv

The `other_signals` file isn't central for the mechanical PIM, but it is important for:

1. **Sanity checks:**
2. Compare cumulative Bayport capex (sum of  $I_{\text{total\_RMB}_y}$ ) with:
  - Registered capital in NIS converted to RMB.
  - EPC contract value and cumulative payments to Masada (USD → RMB).
  - NIS monetary exposures related to Bayport.

3. Check that the Bayport gross stock is of the right order of magnitude relative to these numbers.

**4. Choosing service life and depreciation assumptions:**

5. Use concession end year (e.g., 2046) and project completion year (2021/2022) to get a plausible remaining economic life (e.g., 24–25 years, or longer if you assume some residual value at concession end).

6. Alternatively, align `L_bayport` with whatever you assumed for IPC's main concession assets.

**7. Timings and calibration:**

8. If the notes mention the exact month/year that Bayport began commercial operations, use that to set `dep_start_month`.

9. If there are hints of very front-loaded or back-loaded spending in specific years, you might adjust the within-year investment profile accordingly.

For the other GPT chat, the instructions should be:

- **Load** `other_signals` **as context** but **do not rely on it** to mechanically re-compute K. Use it to:
- Assert approximate consistency (e.g., not off by a factor of 10).
- Document the assumptions used in the final K series (service life, start date, whether cap-interest included, etc.).

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## 8. Checklist of external inputs and assumptions needed

To make the construction fully explicit, here is what the **other chat** must either assume or look up:

1. **FX data:** `fx_rmb_nis.tsv` with annual average NIS per RMB for 2018–2024.
2. **Depreciation assumptions:**
3. `L_bayport` (economic life in years) for Bayport infrastructure.
4.  $\delta_{\text{annual}} = 1 / \text{L\_bayport}$ .
5. Monthly rate  $\delta_{\text{month}} = 1 - (1 - \delta_{\text{annual}})^{(1/12)}$ .
6. **Depreciation start year / month:**
7. `year_dep_start` (e.g., 2022) and `dep_start_month` (e.g., 2022-01).
8. **Definition of productive investment:**
9. Decide whether `I_productive_y` = `T_total_kNIS_y` (completed works only) or `I_total_kNIS_y` (all spending).
10. **Inclusion of capitalized interest:**
11. Decide whether capitalized interest is included in `Gross investment` (recommended) and in the capital stock base.
12. **Treatment of CIP in net stock:**
13. Decide whether to define `K_gross` as `K_completed + CIP` or only `K_completed`, and how this maps into `K_net`.
14. **Within-year investment profile:**

15. Start with uniform monthly distribution; optionally tweak to match any known milestone timing.

As long as the other chat **documents these choices explicitly**, you can later revisit any of them without changing the basic pipeline.

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## 9. Summary of the pipeline (for the other GPT chat)

For quick reference, here is a compact step-by-step pipeline the other GPT chat should follow:

### 1. Load raw data

2. Load `sipg_haifa_bayport_CIP_RMB.tsv` and `sipg_haifa_bayport_other_signals.tsv`.

### 3. Construct annual RMB series

- 4. Filter to Bayport-related rows (`main_Bayport_CIP` + `Bayport_fire_cannon_CIP`).
- 5. For each year, extract `cip_open_rmb`, `cip_close_rmb`, `cip_additions_rmb`, `cap_int_rmb`, `transfers_to_PPE_rmb`, `transfers_to_intangibles_rmb`, `other_decreases_rmb`.
- 6. Compute annual construction spending `I_total_RMB_y` and completed works `T_total_RMB_y`.
- 7. Compute cumulative `K_completed_RMB_y` and `K_gross_RMB_y`.

### 8. Add FX and convert to NIS

9. Load `fx_rmb_nis.tsv`.

10. Merge on `year` and convert all relevant series to NIS, then to kNIS.

### 11. Run annual PIM to define net PPE and depreciation

- 12. Choose `L_bayport`, derive `δannual`.
- 13. Choose `year_dep_start`.
- 14. Define `I_productive_y` (e.g., `T_total_kNIS_y`).
- 15. Run annual PIM to get `K_net_kNIS_y`.
- 16. Compute annual `Depreciation_y` and `Disposals_y` (likely 0).
- 17. Write out `bayport_K_annual.tsv` and use it to fill Panel C.

### 18. Run monthly PIM

- 19. Compute `δmonth`.
- 20. For each year y, distribute `Gross_investment_y` across 12 months (initially uniform).
- 21. Decide construction start month (2018-01) and depreciation start month.
- 22. Run monthly PIM to get `Kt`, `It`, `Dept`, `Dt`.

23. Optionally rescale `I_t` year-by-year to ensure December `K_t` matches annual `K_net_kNIS_y` exactly.

**24. Export monthly series**

25. Write `bayport_K_monthly.tsv` with one row per month and columns described above.

26. **Use** `other_signals` **for validation and documentation**

27. Cross-check magnitudes and ensure no obvious scale errors.

28. Document assumptions (service life, FX choice, dep start, etc.).

This completes the conceptual and technical roadmap for turning your SIPG Bayport raw data into a monthly K-series compatible with the IPC and HPC series.